

Safe Multimodal Decompensation Forecasting in Intensive Care: Joint Modeling of High-Frequency MIMIC-IV Telemetry and Unstructured Clinical Notes with Selective Deferral

Introduction

In intensive care units (ICUs), clinical decision support systems (CDSS) powered by machine learning offer significant potential for early warning of acute physiological decompensation, severe multi-organ dysfunction, and 30-day mortality^{1,11}. Early identification of deteriorating patients allows care teams to deploy rapid response interventions, titrate vasopressors, adjust mechanical ventilation, and initiate fluid resuscitation prior to irreversible systemic collapse^{1,12}. However, existing predictive models deployed in high-acuity environments suffer from two structural vulnerabilities that severely limit their real-world safety and clinical adoption⁴.

The first vulnerability stems from single-modality isolation¹. Traditional clinical machine learning systems typically operate exclusively on structured, numeric bedside telemetry (such as hourly heart rate, mean arterial pressure, and laboratory values) or, less frequently, on periodic unstructured clinical progress notes². Neither data stream in isolation provides a complete representation of patient state¹. Bedside telemetry offers continuous monitoring but lacks qualitative context, diagnostic reasoning, and subjective clinical impressions recorded by nursing staff and physicians². Conversely, clinical notes capture holistic assessments but suffer from documentation delays, irregular recording intervals, and narrative variability². Relying on a single modality creates critical blind spots where physiological anomalies visible in vital sign trajectories are missed due to a lack of clinical context, or subtle narrative warnings are ignored due to absent real-time vital mapping².

The second vulnerability arises from forced-choice inference under severe uncertainty⁴. Conventional deep neural networks emit point-estimate predictions for every patient regardless of data quality, missing telemetry grids, severe artifactual noise, out-of-distribution clinical trajectories, or high predictive ambiguity⁴. In high-imbalance medical domains where positive decompensation events constitute less than 12% of the patient population, uncalibrated deep learning models exhibit dangerous overconfidence⁴. Uncalibrated false negative predictions lead to silent clinical failures, where deteriorating patients with low predicted risk unexpectedly collapse and are denied timely resuscitation⁵. Conversely, uncalibrated false positive predictions generate false alarms that trigger widespread alarm fatigue among clinical staff,

leading to desensitization and ignored alert protocols⁵.

To resolve these dual challenges, this project presents Multimodal EHR Fusion with Selective Clinical Deferral (MM-SCD), a safety-centric deep learning framework designed for acute decompensation

forecasting using the MIMIC-IV database ($N = 75,672$ patient stays)¹. The system integrates a dual-branch neural architecture: a 2-layer Bidirectional Long Short-Term Memory (Bi-LSTM) network processes regularized 24-hour physiological time-series and explicit binary missingness indicators, while a Bio_ClinicalBERT encoder captures fine-grained semantic representations from 24-hour unstructured clinical progress notes⁶. These representations are dynamically unified using a Multi-Head Cross-Attention Fusion layer, allowing temporal vital sign states to selectively query and weight pertinent narrative text tokens⁶.

Crucially, rather than enforcing binary predictions on ambiguous clinical trajectories, MM-SCD incorporates an information-theoretic selective deferral gate⁴. By calculating normalized Shannon

Predictive Entropy $H(X)$, the system quantifies predictive uncertainty per patient case⁴. When

predictive entropy exceeds an operational safety threshold ($H(X) > \gamma$), the system abstains from automated prediction and defers the decision to human clinical Rapid Response Teams (RRT)⁴. This report synthesizes the complete technical strategy, execution pipeline, empirical research findings, and clinical evaluation generated across project artifacts.

Related Work

To establish the context of our contributions, this section examines key published research projects that addressed clinical prediction, multimodal fusion, uncertainty quantification, and learning to defer in intensive care environments, detailing the specific problems they tackled and how their methodologies operated¹.

Project 1: Joint Time-Series and Clinical Note Modeling for ICU Tasks (Khadanga et al., 2019)

- **Problem Addressed:** Standard clinical decision support systems struggled to combine regularly sampled bedside telemetry with intermittently charted, unstructured physician notes¹. Prior benchmarking efforts evaluated time-series in isolation, missing crucial diagnostic insights recorded in physician progress narratives¹.
- **How They Dealt With the Problem:** Khadanga et al. developed a multimodal deep neural network that jointly models continuous vital signs and discrete textual note events on MIMIC benchmark tasks (in-hospital mortality, decompensation, and length-of-stay)¹. To tackle asynchronous data streams, they utilized a Recurrent Neural Network (LSTM) to process 1-hour temporal vital sign vectors while deploying a 1D Convolutional Neural Network (CNN) to extract

local n-gram semantic features from physician progress notes recorded at irregular intervals¹. The temporal hidden states of the LSTM and the text representations from the CNN were concatenated into a unified feature vector to drive classification heads⁶.

- **Limitations and Contrast with MM-SCD:** While Khadanga et al. proved that adding clinical notes improves prediction accuracy across ICU benchmark tasks¹, their architecture relied on static feature concatenation and basic 1D-CNN text embeddings, which fail to capture deep contextual semantics or dynamic, token-level cross-modal attention⁶. Furthermore, their system enforced predictions on all patient records without evaluating model uncertainty or offering abstention mechanisms⁴.

Project 2: Multimodal Decomensation Modeling and Transformer Cross-Attention (MDPI Algorithms 2024; Deznabi et al., 2021; Zhang et al., 2022)

- **Problem Addressed:** Unstructured clinical text and time-series telemetry contain complex non-linear correlations and asynchronous sampling patterns that simple additive or concatenation fusion strategies fail to align³.
- **How They Dealt With the Problem:** Recent multimodal studies expanded upon early fusion by integrating sequence architectures (LSTMs/RNNs) with pre-trained BERT/BioBERT encoders and cross-attention fusion modules⁶. For example, studies on sequence-to-sequence ICU decomensation prediction utilized CNN-LSTM networks on patient vitals and clinical characteristics to forecast risk sequences across dynamic observation windows³. Similarly, cross-attention architectures were introduced to allow time-series hidden states to directly query textual representations, capturing fine-grained dependencies between continuous vital sign shifts and narrative clinical notes⁶.
- **Limitations and Contrast with MM-SCD:** Existing cross-attention multimodal architectures focus almost exclusively on improving point-estimate classification metrics (such as AUROC/AUPRC)³. However, they emit uncalibrated predictions even when processing incomplete telemetry grids or noisy text, lacking real-time uncertainty quantification to defer ambiguous patient stays to clinical specialists⁴.

Project 3: Learning to Defer with Uncertainty (LDU) in Clinical Diagnosis (Mozannar & Sontag, 2020; PMC8810991)

- **Problem Addressed:** Deep neural networks deployed in clinical decision-making are prone to uncalibrated overconfidence, yielding silent false negative failures on high-risk, uncertain patient trajectories where machine predictions should be reviewed by human experts⁴.
- **How They Dealt With the Problem:** The authors formulated the Learning to Defer with Uncertainty (LDU) algorithm, which identifies patients for whom diagnostic uncertainty is high

and defers them for evaluation by human medical experts⁴. The methodology operates in two distinct stages⁴:

1. *Uncertainty Quantification*: Stage 1 trains a Deep Ensemble of K neural networks under varied random initializations⁴. The individual model probability outputs $(M_1(x_i), M_2(x_i), \dots, M_k(x_i))$ are aggregated to calculate Diagnostic Entropy $u(x)$, isolating predictive uncertainty⁴.
2. *Deferral Network Optimization*: Stage 2 trains a specialized 'deferral' neural network that takes as input the ensemble output probabilities and entropy values ($K + 2$ inputs total)⁴. The network uses a cost-sensitive loss function combining classification cross-entropy and a weighted defer loss controlled by parameter α ⁴:

$$\mathcal{L}_{LDU} = \mathcal{L}_{\text{class}}(y, \hat{y}) + \alpha \cdot \mathcal{L}_{\text{defer}}(d)$$

By decreasing α , the network is encouraged to defer ambiguous cases to human evaluation, learning an explicit decision function $f(x) \in \{1, \dots, C, \text{defer}\}$ to optimize overall system accuracy⁴.

- **Limitations and Contrast with MM-SCD**: While LDU established the utility of entropy-guided abstention, training a K -model deep ensemble requires high computational resources and inference latency, making real-time deployment challenging in high-throughput ICUs⁴. MM-SCD builds upon LDU's theoretical foundation by applying single-forward-pass normalized Shannon Predictive Entropy $H(X)$ directly onto cross-attended multimodal representations, delivering a computationally efficient selective deferral system tailored for MIMIC-IV decompensation forecasting⁴.

Project 4: Visualized Learning and Per-Patient Uncertainty Quantification (VL4ML; arXiv:2205.04599)

- **Problem Addressed**: Standard ML models present predictions as abstract point probabilities without exposing confidence boundaries, making it difficult for clinicians to identify out-of-distribution cases, bias, or erroneous classifications during ICU survival prediction tasks⁵.
- **How They Dealt With the Problem**: The authors introduced Visualized Learning for Machine Learning (VL4ML), an explicit human-centered explainability paradigm evaluating 24-hour EHR

data (e.g., WiDS Datathon challenge dataset with 183 tabular features)⁵. VL4ML maps complex feature vector representations into intuitive spatial target images where central regions indicate classification outcomes (survival vs. fatal warning) and surrounding reserved areas symbolize Reserved Uncertainty (RU) boundaries⁵. Per-patient uncertainty quantification allows the algorithm to recognize ambiguous cases and abstain from making forced automated predictions, facilitating human-machine decision sharing⁵.

- **Limitations and Contrast with MM-SCD:** VL4ML focuses primarily on static tabular features and visual representations⁵. MM-SCD extends the principle of per-patient uncertainty quantification directly into deep multimodal time-series and transformer architectures using mathematical entropy thresholds to automate low-risk cases while routing ambiguous cases to Rapid Response Teams⁴.

Project 5: Continuous Physiological Waveforms & Narrative Risk Analytics (VitalML; PMC10073111; PMC9825202)

- **Problem Addressed:** Predictive modeling in critical care often ignores either raw continuous waveform signals or the rich contextual narratives recorded in free-text clinical notes, leading to delayed identification of patient deterioration⁷.
- **How They Dealt With the Problem:** Studies such as VitalML combined triage data with features extracted from continuous ECG/PPG waveforms via conventional signal processing and transformer embeddings to predict clinical decompensation (tachycardia, hypotension, hypoxia)¹⁰. Concurrently, studies evaluating home health and ICU environments demonstrated that NLP features extracted from free-text clinical progress notes capture socio-behavioral and qualitative risk indicators that significantly enhance predictive power over structured data alone⁷.
- **Limitations and Contrast with MM-SCD:** VitalML and clinical note risk frameworks operate on isolated data modalities or engineered static feature sets⁷. MM-SCD unifies high-frequency temporal telemetry grids and unstructured Bio_ClinicalBERT narrative embeddings within an end-to-end Multi-Head Cross-Attention network, coupled with real-time entropy-based deferral⁴.

System Architecture and Experimental Methods

This section details cohort extraction, multimodal preprocessing, dynamic SOFA target construction, the dual-branch neural architecture, Focal Loss optimization, and the information-theoretic selective deferral mechanism¹.

Cohort Selection and Multimodal Preprocessing

Patient records were extracted from the MIMIC-IV database (v3.1), specifically linking icustays, patients, chartevents, labevents, and discharge note modules¹. The study cohort was defined by selecting adult

patients ($\text{anchor_age} \geq 18$) with an ICU length of stay of at least 24 hours¹. For each ICU stay, data was aggregated over the initial 24-hour window following admission ($t \in [0, 24]$ hours), resulting in $N = 75,672$ processed stay records.

The continuous physiological telemetry comprises eight key vital signs and laboratory markers: Heart Rate (itemid 220045), Mean Arterial Pressure (MAP, itemid 220052), Respiratory Rate (itemid 220210), Oxygen Saturation (SpO_2 , itemid 220277), Body Temperature (itemid 223761), Blood Lactate (itemid 50813), White Blood Cell count (WBC, itemid 50912), and Serum Creatinine (itemid 51301)². The 24-hour timeline was discretized into 1-hour temporal bins ($T = 24$)¹. To account for severe missingness inherent in EHR telemetry, a dual-channel grid representation $X_{ts} \in \mathbb{R}^{24 \times 16}$ was engineered². Channel 1 contains Z-score normalized vital sign values normalized against training set global means and standard deviations². Missing entries were imputed using forward-fill followed by zero-imputation². Channel 2 consists of a binary missingness mask $M \in \{0, 1\}^{24 \times 8}$, where $M_{t,f} = 1$ indicates a true physiological measurement at hour t for feature f , and $M_{t,f} = 0$ indicates an imputed value².

Unstructured clinical text recorded during the initial 24 hours was concatenated and tokenized using the Hugging Face Bio_ClinicalBERT tokenizer (emilyalsentzer/Bio_ClinicalBERT) with a maximum sequence truncation length of $L = 256$ tokens⁶.

Dynamic SOFA Trajectory Targets

Clinical target labels capture severe physiological decompensation and 30-day post-ICU mortality¹. To ground target labels in objective organ dysfunction, hourly Sequential Organ Failure Assessment (SOFA) sub-scores were computed dynamically across the 24-hour window³:

- Cardiovascular Component: $\text{MAP} < 70 \text{ mmHg}$ (+1 point).
- Respiratory Component: $\text{SpO}_2 < 96\%$ (+1 point), $< 92\%$ (+2 points), $< 88\%$ (+3 points).
- Renal Component: $\text{Serum Creatinine} \geq 1.2 \text{ mg/dL}$ (+1 point), ≥ 2.0 (+2 points), ≥ 3.5 (+3 points), ≥ 5.0 (+4 points).

A binary decompensation target $Y \in \{0, 1\}$ was assigned if a patient experienced 30-day mortality

or exhibited severe organ dysfunction defined by a peak 24-hour SOFA score $\text{SOFA}_{\max, 24h} \geq 4$ ³. The overall dataset exhibits severe positive class imbalance, with positive decompensation events representing approximately 12% of stays.

Dual-Branch Cross-Attention Neural Architecture

The MM-SCD model consists of two specialized feature encoders coupled through a Multi-Head Cross-Attention fusion module⁶:

1. **Time-Series Bi-LSTM Encoder:** The dual-channel input tensor $X_{ts} \in \mathbb{R}^{24 \times 16}$ passes through a 2-layer Bidirectional LSTM network¹. The Bi-LSTM processes forward and backward temporal sequences, generating hidden state sequences $H_{ts} \in \mathbb{R}^{24 \times D}$ (where $D = 256$)². The final temporal hidden state vector $h_{ts} \in \mathbb{R}^D$ summarizes the 24-hour physiological trajectory².
2. **Textual Bio_ClinicalBERT Encoder:** Tokenized clinical notes $(I_{\text{ids}}, I_{\text{mask}})$ pass through Bio_ClinicalBERT⁶. To ensure training stability and prevent over-fitting while preserving contextual domain knowledge, lower Transformer layers are frozen, while the top two encoder layers are un-frozen for differential fine-tuning⁶. Token embeddings pass through a linear projection layer, Layer Normalization, and Dropout ($p = 0.2$), yielding textual token representations $H_{\text{text}} \in \mathbb{R}^{L \times D}$ ⁶.
3. **Multi-Head Cross-Attention Fusion:** Rather than concatenating static vectors, MM-SCD utilizes cross-attention where the temporal vital sign representation h_{ts} acts as a Query ($Q \in \mathbb{R}^{1 \times D}$), while narrative text token embeddings H_{text} act as Keys ($K \in \mathbb{R}^{L \times D}$) and Values ($V \in \mathbb{R}^{L \times D}$)⁶. Multi-head attention with $N_h = 4$ heads computes scaled dot-product attention weights:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

Explicit key padding masking is applied to ignore zero-padded text tokens⁶. A residual connection fuses the temporal state h_{ts} with the cross-attended text context, followed by Layer Normalization:

$$c_{\text{fused}} = \text{LayerNorm}(h_{ts} + \text{CrossAttention}(h_{ts}, H_{\text{text}}, H_{\text{text}}))$$

The fused vector $z = [h_{ts} \parallel c_{\text{fused}}] \in \mathbb{R}^{2D}$ is fed into a multi-layer classification head with Batch Normalization, ReLU activation, Dropout ($p = 0.3$), and a final linear layer emitting raw logits⁶.

Loss Function and Optimization

To address extreme class imbalance, the network is trained using Focal Loss:

$$\mathcal{L}_{\text{Focal}}(p_t) = -\alpha(1 - p_t)^\gamma \log(p_t)$$

where p_t represents the model's estimated probability for the true class, $\alpha = 0.25$ balances positive/negative class weighting, and $\gamma = 2.0$ dynamically down-weights easy, well-classified examples while focusing gradients on hard misclassified decompensations. Optimization uses AdamW with differential learning rates (2×10^{-5} for Bio_ClinicalBERT layers, 1×10^{-3} for attention and classifier heads), gradient norm clipping at 1.0 , and Cosine Annealing learning rate scheduling over 4 epochs.

Selective Deferral Mechanism and Cost-Aware System Loss

Predictive uncertainty is quantified per patient using normalized Shannon Predictive Entropy $H(X)$ expressed in bits⁴:

$$H(X) = - \sum_{i \in \{0,1\}} P(Y = i \mid X) \log_2 P(Y = i \mid X)$$

For an automated target coverage rate $\Phi \in (0, 1]$, an operating threshold γ_Φ is computed from the empirical percentile of test set entropies⁴. Samples with $H(X) \leq \gamma_\Phi$ are classified automatically by the neural network, whereas samples with $H(X) > \gamma_\Phi$ are deferred to human clinical RRT

evaluation⁴.

To evaluate the economic and clinical operational trade-offs of deferral, Cost-Aware System Loss

$L_{sys}(\Phi)$ is defined⁴:

$$L_{sys}(\Phi) = (R_{sel}(\Phi) \cdot \Phi) + (c_{defer} \cdot (1 - \Phi))$$

where $R_{sel}(\Phi)$ represents selective empirical classification risk on automated cases, and c_{defer} represents the fixed unit cost of clinical RRT review relative to automated misclassification error⁴.

The end-to-end operational workflow of data extraction, dual-branch feature encoding, cross-attention alignment, loss optimization, and entropy-gated selective deferral is summarized in Table 1 below¹:

Workflow Stage	Input Components	Architectural Mechanics & Operations	Output Deliverables
1. Cohort & Data Ingestion	MIMIC-IV icustays, chartevents, labevents, discharge notes	Query 24h temporal telemetry and discharge notes for ICU stays $\geq 24h$; discretize vitals into $T=24$ bins ¹ .	Filtered patient cohort metadata (N=75,672 patient stay records).
2. Dual-Modality Preprocessing	8 vital/lab items + unstructured progress notes	Impute telemetry via forward-fill; generate Z-scores and binary observation masks M; tokenize text via Bio_ClinicalBERT (L=256) ² .	Dual-channel grid X_{ts} in $R^{24 \times 16}$ and token tensors (input_ids, attention_mask).
3. Temporal & Textual Encoding	Dual-channel time-series grid + tokenized text notes	2-layer Bi-LSTM extracts temporal vector h_{ts} in R^{256} ;	Temporal trajectory vector h_{ts} and contextual text representations

		fine-tuned Bio_ClinicalBERT extracts token embeddings H_{text} in $R^{256 \times 256}$.	H_{text} .
4. Multi-Head Cross-Attention	Vectors h_{ts} (Query) and H_{text} (Key, Value)	4-head cross-attention allows temporal vitals to query text tokens; residual connection and LayerNorm yield fused vector z in R^{512} .	Dynamic cross-attention representation z and token alignment matrix A .
5. Classification & Focal Optimization	Fused vector z in R^{512} + true label Y in $\{0, 1\}$	Dense classification head with BatchNorm & Dropout ($p=0.3$); Focal Loss ($\alpha=0.25$, $\gamma=2.0$) optimization via AdamW.	Decompensation logits, class probabilities $P(Y)$
6. Uncertainty Gating & Deferral	Softmax class probabilities $P(Y)$	X	Calculate Shannon Predictive Entropy $H(X)$ in bits; compare against operating threshold γ_{ϕ} .

Table 1: End-to-end technical workflow and architectural pipeline for MM-SCD.

Empirical Findings and System Evaluation

The empirical pipeline evaluated five distinct model configurations on the processed MIMIC-IV cohort ($N = 75,672$): a pure PyTorch Tabular Baseline (TabularMLP), a Time-Series Only Bi-LSTM (ts_only), a Bio_ClinicalBERT Text Only model (text_only), a Concatenation Multimodal model (concat), and the Cross-Attention Multimodal model (multimodal). All deep learning models were trained for 4 epochs (3,311 steps/epoch).

The exact quantitative evaluation results obtained from the pipeline execution logs are presented in Table 2 below:

Model Architecture	Validation AUROC	Train Loss (Epoch 4)	Validation Loss (Epoch 4)	Epoch Time (s)	Best Checkpoint Val AUROC
Tabular MLP Baseline (TabularMLP)	0.8061	--	--	--	0.8061
Time-Series Only Bi-LSTM (ts_only)	0.7596	0.0248	0.0243	~79.1 s	0.7596
ClinicalBERT Text Only (text_only)	0.5177	0.0281	0.0303	~437.3 s	0.5177
Concat Multimodal (concat)	0.7544	0.0250	0.0245	~449.4 s	0.7544
MM-SCD Cross-Attention (multimodal)	0.7538	0.0248	0.0247	~455.2 s	0.7538

Table 2: Exact empirical performance across baseline and deep learning model modes from notebook execution logs.

Additional Metrics for PyTorch Tabular Baseline

- **AUROC:** 0.8061
- **AUPRC:** 0.4407

- **Expected Calibration Error (ECE):** 0.0113
- **Brier Score:** 0.1047

Detailed Analysis of Empirical Findings and Training Dynamics

1. Superior Performance of the Tabular MLP Baseline:

The PyTorch Tabular MLP baseline achieved the highest discriminative performance across all evaluated models with a Validation AUROC of 0.8061, an AUPRC of 0.4407, and superior calibration (ECE 0.0113). Aggregating 24-hour summary statistics (means, standard deviations, minimums, maximums) acts as an effective low-pass filter over temporal noise and missingness artifacts, allowing a dense MLP to optimize efficiently without complex sequence parameterization.

2. Time-Series Bi-LSTM Convergence:

The `ts_only` Bi-LSTM model reached a Validation AUROC of 0.7596 at Epoch 4 (Train Loss: 0.0248, Val Loss: 0.0243). Fast epoch execution times (~79 seconds) enabled steady epoch-over-epoch AUROC improvements (0.7243 in Epoch 1 $\rightarrow$ 0.7356 in Epoch 2 $\rightarrow$ 0.7592 in Epoch 3 $\rightarrow$ 0.7596 in Epoch 4), demonstrating stable convergence on structured telemetry.

3. Under-Convergence of the Transformer Text Branch (`text_only`):

The `text_only` Bio_ClinicalBERT model achieved a Validation AUROC of 0.5177 after 4 epochs (437 seconds/epoch, totaling ~29 minutes of training time). Because Transformer language models possess over 110 million parameters, 4 epochs are insufficient for lower-layer fine-tuning on noisy 24-hour ICU notes, leaving textual representations near chance performance (0.5177).

4. Impact on Multimodal Fusion (`concat` and `multimodal`):

Due to the under-convergence of the textual encoder, the combined multimodal modes (`concat` at 0.7544 Val AUROC and `multimodal` cross-attention at 0.7538 Val AUROC) achieved performance very close to the `ts_only` baseline (0.7596). The un-converged text embeddings act as a noisy input to the fusion layer during early training epochs. However, both multimodal models showed consistent positive trajectories across epochs (multimodal: 0.6936 in Epoch 1 $\rightarrow$ 0.7145 in Epoch 2 $\rightarrow$ 0.7479 in Epoch 3 $\rightarrow$ 0.7538 in Epoch 4), indicating that discriminative capacity continues to climb as training progresses.

Analysis of False Alarms vs. Silent False Negatives in ICU Triage

Clinical decision support systems in ICUs face two distinct failure modes⁵:

- **Silent False Negatives (High Clinical Risk):** Cases where a patient has a low estimated probability of dying or decompensating but unexpectedly deteriorates and dies (analogous to Case (h) in the VL4ML study)⁵. In uncalibrated point-prediction models, these false negatives cause

clinicians to withhold timely resuscitation⁵.

- **False Alarms / False Positives (Resource Strain):** Cases where a patient is estimated to have a high probability of decompensation but survives (analogous to Case (g) in the VL4ML study)⁵. Uncalibrated overprediction causes alarm fatigue among nursing staff⁵.

Applying normalized Shannon Predictive Entropy $H(X)$ to model output probabilities addresses both failure modes⁴. Setting an automated coverage threshold at $\Phi = 0.80$ defers the top 20% highest-entropy cases (which contain the highest concentration of boundary false negatives and false positives) to human Rapid Response Teams, pruning high-uncertainty predictions and lowering retained empirical risk⁴.

Subgroup Fairness Audit across Gender Cohorts

Evaluating model outputs across demographic subsets confirmed equitable performance across male ('M') and female ('F') cohorts. Validation AUROC and deferral rates remained highly consistent across groups without introducing demographic performance disparities.

Conclusion and Future Directions

This study benchmarked PyTorch Tabular and deep multimodal architectures for 24-hour ICU acute decompensation forecasting on MIMIC-IV ($N = 75,672$). The empirical findings reveal that under short 4-epoch fine-tuning schedules, the PyTorch Tabular MLP baseline achieves superior performance (AUROC 0.8061, AUPRC 0.4407, ECE 0.0113) compared to deep time-series (0.7596) and multimodal cross-attention architectures (0.7538).

Key insights and future research directions include:

- **Extended Multimodal Fine-Tuning Schedule:** The primary bottleneck for multimodal performance is textual under-convergence (Bio_ClinicalBERT reached 0.5177 AUROC at Epoch 4)⁶. Given that each multimodal epoch requires ~455 seconds (~7.5 minutes), future work with expanded GPU time budgets should extend training to 15–20 epochs with dynamic layer unfreezing and warm-up scheduling to allow the cross-attention mechanism to surpass the tabular baseline⁶.
- **Feature-Engineered Tabular Hybrids:** Given the strong performance of TabularMLP (0.8061 AUROC), hybrid models that feed tabular summary features directly into the cross-attention fusion head alongside Bi-LSTM states and BERT embeddings can combine the stability of summary stats with dynamic text semantics⁵.
- **Production-Grade Selective Deferral with LDU Parameters:** Integrating single-forward-pass

Shannon entropy gating alongside cost-sensitive deferral parameters (α) onto top-performing models ensures that high-uncertainty ICU stays are routed to clinical Rapid Response Teams, preventing silent false-negative failures in high-acuity patient care⁴.

- **Expansion to Multi-Modal Imaging and Waveforms:** Extending the architecture to incorporate chest X-rays (CXR) via vision transformers and high-frequency continuous ECG waveforms to establish tri-modal clinical fusion¹³.
- **Dynamic Continuous Streaming Inference:** Transitioning from a 24-hour static observation window to real-time streaming inference using time-decaying attention kernels to provide continuously updated decompensation risk trajectories at the bedside.
- **Multi-Center Validation and Lightweight Bayesian LLMs:** Testing cross-hospital domain generalization on external databases such as eICU-CRD, while integrating lightweight Bayesian uncertainty quantification mechanisms (such as MedBayes-Lite) to further refine epistemic and aleatoric uncertainty estimation in multimodal clinical AI¹⁴.

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